

Hands-on Lab 2: Creating Advanced Charts

In this lab, you will learn how to create some advanced charts in Excel. We'll first create a sunburst chart, then a scatter chart, and lastly a histogram.

Software Used in this Lab

The instruction videos in this course use the full Excel Desktop version as this has all the available product features, but for the hands-on labs we will be using the free 'Excel for the web' version as this is available to everyone.

Although you can use the Excel Desktop software if you have access to this version, it is recommended that you use Excel for the web for the hands-on labs as the lab instructions specifically refer to this version, and there are some small differences in the interface and available features. If you do not yet have access to Excel for the Web, you can follow the instructions in the following lab to get started with it: [Hands-on Lab: Introduction to Excel for the web](#).

Dataset Used in this Lab

The dataset used in this lab comes from the following source: <https://www.kaggle.com/gagandeep16/car-sales> under a [CC0: Public Domain license](#). We are using a modified subset of that dataset for the lab, so to follow the lab instructions successfully, please use the dataset provided with the lab, rather than the dataset from the original source.

Objectives

After completing this lab, you will be able to:

- Create a hierarchical type chart such as a sunburst chart.
- Create a scatter chart.
- Create a statistical chart such as a histogram.

Exercise 1 : Creating Sunburst, Scatter and Histogram Charts in Excel

In this exercise, you will learn how to create advanced charts, such as a sunburst chart, a scatter chart and histogram charts in Excel.

Task A : Create a Sunburst Chart

1. Download the file [Car_Sales_Kaggle_DV0130EN_Lab2_Start.xlsx](#). Upload and open it using Excel for the web.
2. Switch to the worksheet named **Sunburst Chart**.
3. Click the **drop-down arrow** at the top of column **A (Manufacturer)**.
4. In the list of filters, click **Select All** to DESELECT all filters, then scroll down the filter list and ONLY select **Nissan**, then click **Apply**.
5. Select column **B**, then hold **SHIFT** and select column **C**.
6. On the **Charts** group of the **Insert** tab, click **Other Charts** and choose **Sunburst** from the **Hierarchical** category.

Open in Desktop App



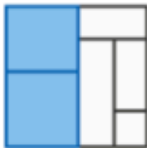
Comment

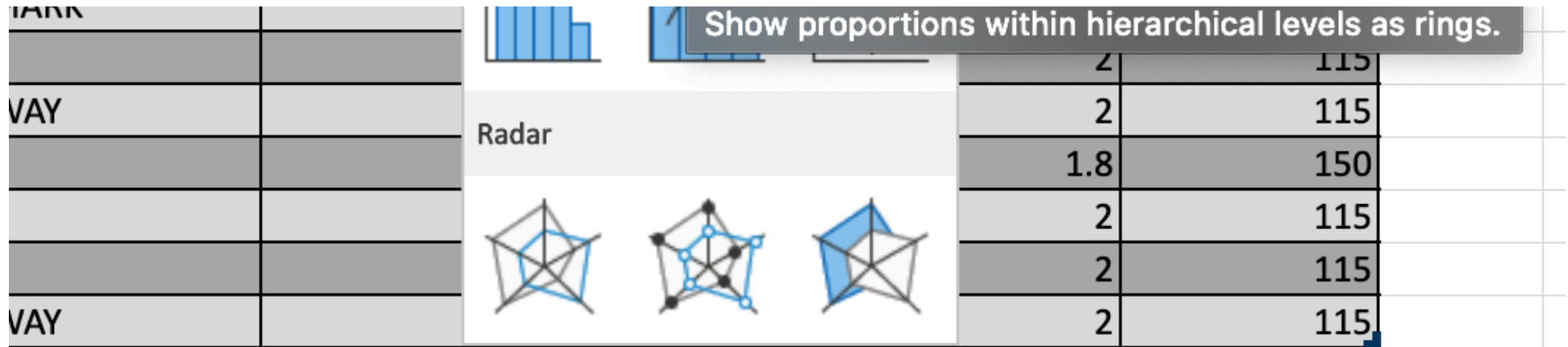
Charts ▾



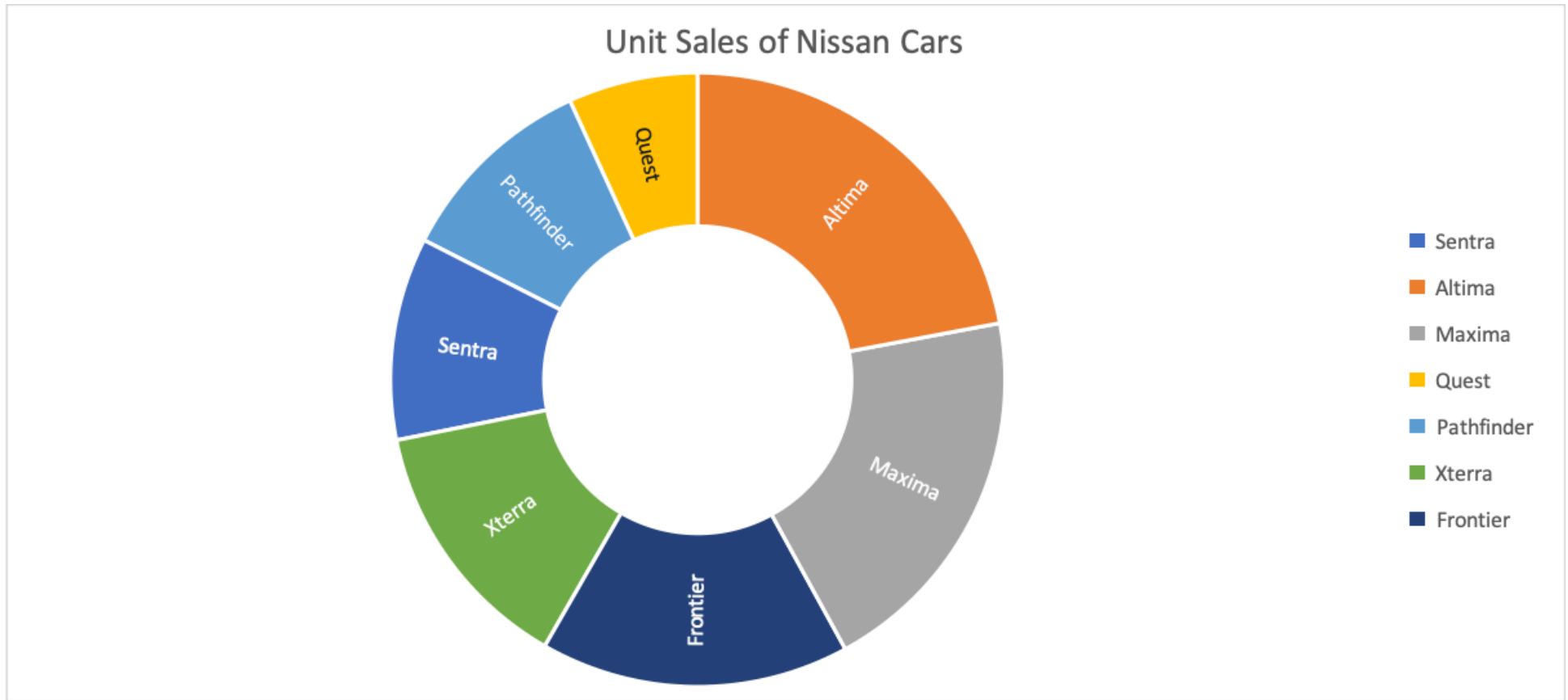
A histogram showing the distribution of the number of children per family. The x-axis is labeled 'Number of children' with values 0, 1, 2, 3, 4, 5. The y-axis is labeled 'Frequency' with values 0, 1, 2, 3, 4, 5. The bars have the following heights: 0 children: 1, 1 child: 2, 2 children: 3, 3 children: 1, 4 children: 0, 5 children: 0.

Compare values across hierarchy levels.

MARK		Waterfall		2.8	200
MARK				4.2	310
				2.5	170
				2.8	193
		Funnel		2.8	193
				2.3	185
CE				3.2	221
				4.3	275
				5	302
DA		Hierarchical		2.3	190
VAY				2.3	185
				3.2	215
EN				5	302
		Statistical			
					
MARK					



7. Click on the floating chart area to access the **Chart** tab in the ribbon.
8. On the **Labels** group of the **Chart** tab, click **Chart Title** and select **Edit Chart Title...**
9. In the text input area of the dialog box **Edit Title**, write “**Unit Sales of Nissan Cars**” and click **OK**.
10. On the **Labels** group of the **Chart** tab, click **Legend** and select **Show Legend at Right**.
11. Your chart should look something like the one below:



Task B : Create a Scatter Chart

1. Switch to the worksheet named **Scatter Chart**.
2. Click the **filter drop-down** in column **A (Manufacturer)**.
3. In the filter list, only select **Audi, BMW, Mercedes-B, Porsche, Volkswagen** and click **OK**.
4. Select column **E**, then hold **SHIFT** and select column **F**.
5. On the **Charts** group of the **Insert** tab, click **Scatter** Chart and choose **Scatter with only Markers** from the **Scatter** category.

Help Design Tell me what you want to do Open in Desktop App

Column Line Pie Bar Area Scatter Other Charts Hyperlink Comment

Charts

ARK	\$33,950		69%	2.8	
ARK	\$62,000		63%	4.2	
	\$26,990		91%	2.5	
	\$33,400				
	\$38,900				
	\$31,750				
E	\$49,900				
	\$69,700				
	\$82,600		71%	5	
A	\$38,900		83%	2.3	
AY	\$41,000	\$28,635	70%	2.3	

Scatter

Scatter with only Markers

Compare pairs of values.

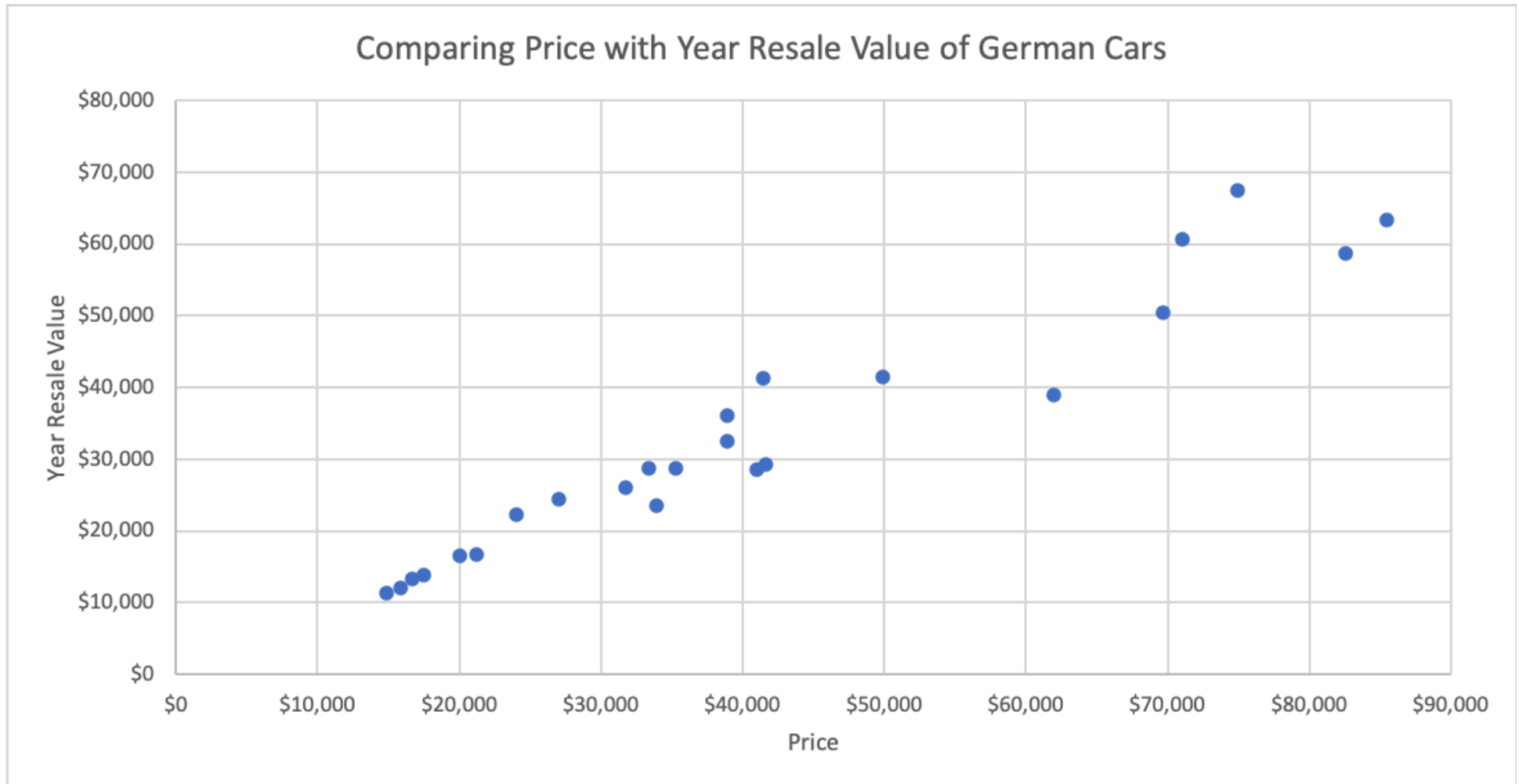
Use it when the values are not in x-axis order or when they represent separate measurements.

6. Drag the chart across to the right of the sheet.

7. Click on the floating chart area to access the **Chart** tab in the ribbon.

8. On the **Labels** group of the **Chart** tab, click **Chart Title** and select **Edit Chart Title....**

9. In the text input area of the dialog box **Edit Title**, write “**Comparing Price with Year Resale Value of German Cars**” and click **OK**.
10. On the **Labels** group of the **Chart** tab, click **Axis Titles** and select **Primary Horizontal Axis Title > Edit Horizontal Axis Title....**
11. In the text input area of the dialog box **Edit Title**, write “**Price**” and click **OK**.
12. On the **Labels** group of the **Chart** tab, click **Axis Titles** and select **Primary Vertical Axis Title > Edit Vertical Axis Title....**
13. In the text input area of the dialog box **Edit Title**, write “**Year Resale Value**” and click **OK**.
14. On the **Labels** group of the **Chart** tab, click **Legend** and select **None**.
15. Your chart should look something like the one below:



Task C : Create a Histogram Chart

1. Switch to the worksheet named **Histogram Chart**.
2. Click the **drop-down arrow** at the top of column **A (Manufacturer)**.
3. In the filter list, only select **Audi, BMW, Mercedes-B, Porsche, Volkswagen** and click **OK**.
4. Select column **B**, then hold **SHIFT** and select column **C**.

5. On the **Charts** group of the **Insert** tab, click **Other Charts** Chart and choose **Histogram** from the **Statistical** category.

Tell me what you want to do

Open in Desktop App



Pie



Bar



Area



Scatter

Other
Charts ▾

Hyperlink



Comment

Charts

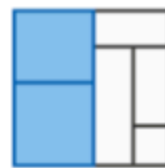
Waterfall



Funnel

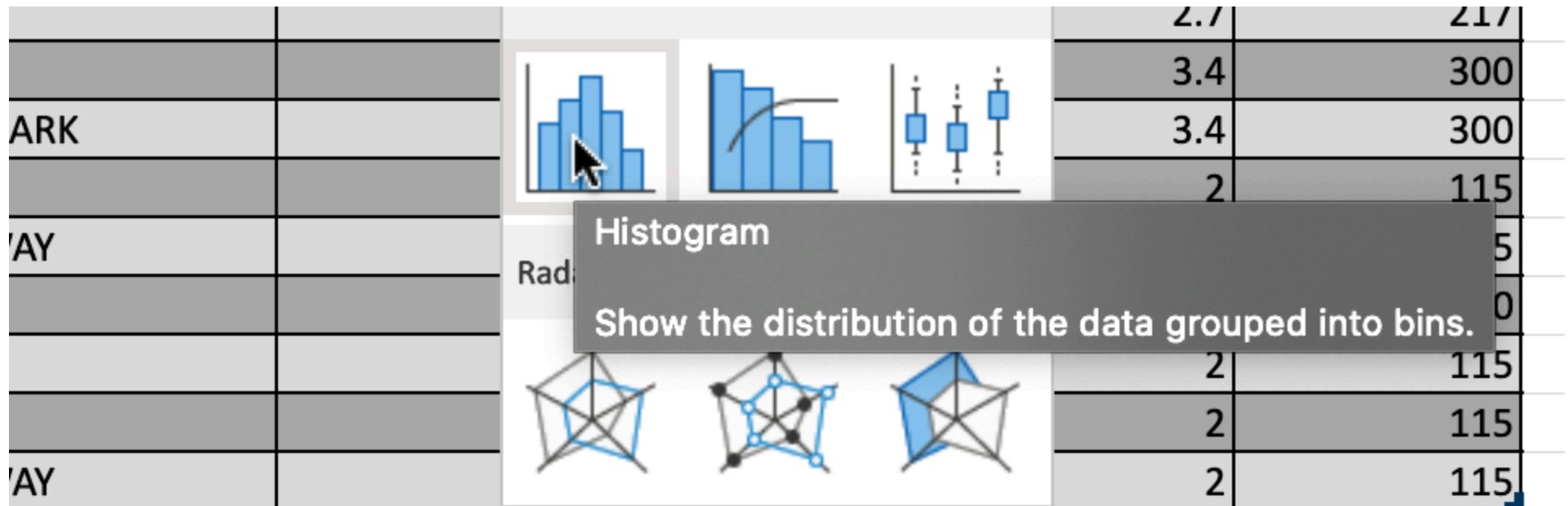


Hierarchical



Statistical

ARK		2.8	200
ARK		4.2	310
		2.5	170
		2.8	193
		2.8	193
		2.3	185
CE		3.2	221
		4.3	275
		5	302
DA		2.3	190
AY		2.3	185
		3.2	215
EN		5	302
		3.2	215
		2.7	217



6. Drag the chart across to the right of the sheet.

7. Click on the floating chart area to access the **Chart** tab in the ribbon.

8. On the **Labels** group of the **Chart** tab, click **Chart Title** and select **Edit Chart Title...**

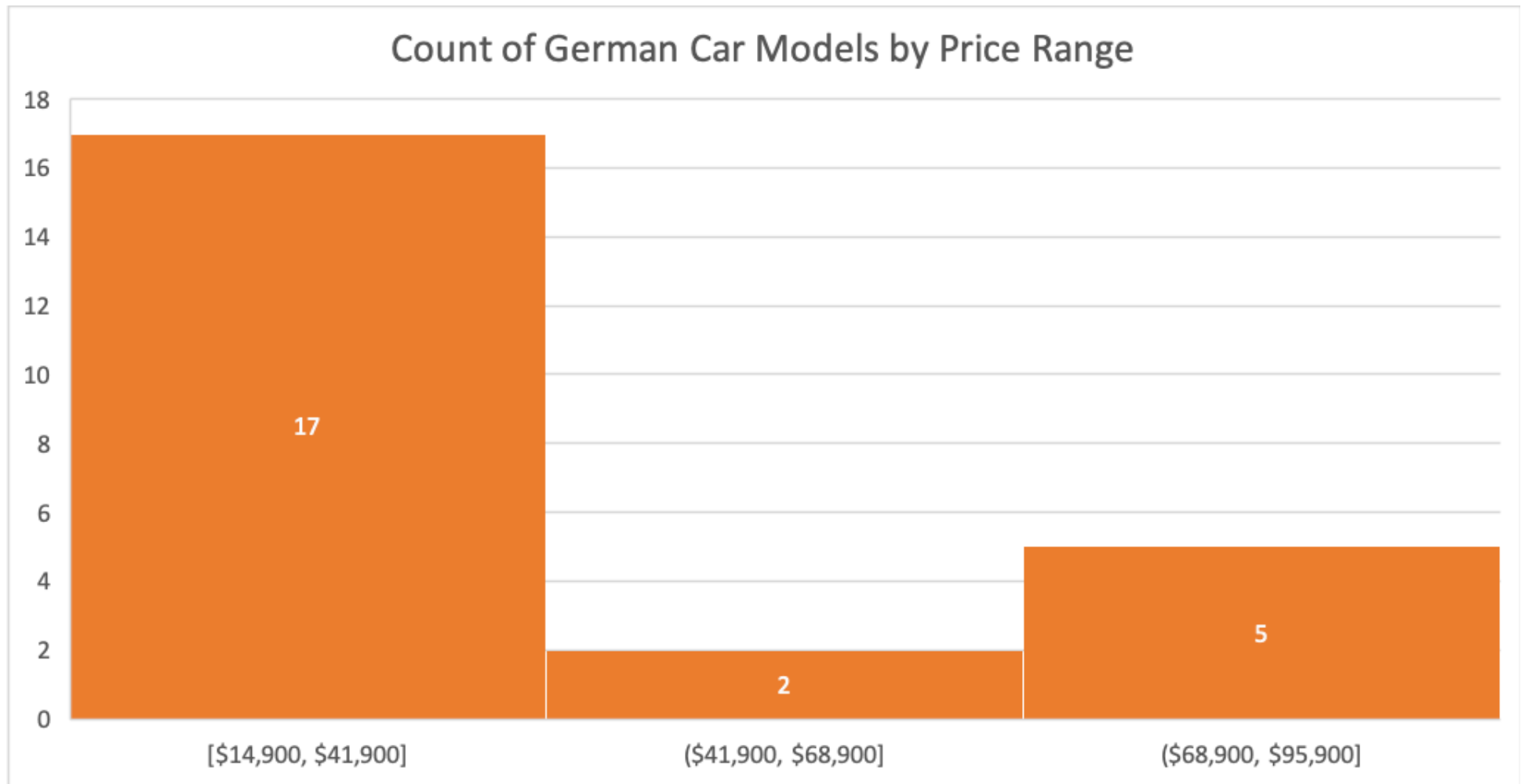
9. In the text input area of the dialog box **Edit Title**, write “**Count of German Car Models by Price Range**” and click **OK**.

10. On the **Labels** group of the **Chart** tab, click **Data Labels** and select **Center**.

11. On the **Format** group of the **Chart** tab, click **Format**.

12. On the right side menu bar **Format**, select **Series “Price” > Fill > Orange, Accent 2**.

13. Your chart should look something like the one below:



Congratulations! You have completed Lab 2, and you are ready for the next topic.